



Upgrading to/ Downgrading from 2022 MUPP Update

11/28/2022

Rev. 1

ABSTRACT:

This document's purpose is to outline the changes required for a MUPP update from the 2012 to 2022 revision and a MUPP rollback from the 2022 to 2012 revision. Both processes require the removal and addition of CMs to the MUPP C300 controller and MODBUS/TCP and Analog I/O changes; as such, this document will describe the procedure for making these changes. As this document covers I/O changes, it can also be consulted for the methodology of making either a MODBUS/TCP or Analog I/O change; however, this is only a secondary function of this document.

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Document Control and Revision History

Table 1: Document Control and Revision History

Revision	Date	Author(s)	Reviewed by	Comments/Details
0	2022	Nicholas O'Keeffe		

Definitions

MUPP	Murdoch University Pilot Plant
CM	Control Module
EPKS	Experion Process Knowledge System
I/O	Input/Output
LUP	Lamella Underflow Pump
FCV	Flow Control Valve
DEVADDR	Device Address
CHANNUM	Channel Number
PCDINUMARRCH	Peer Control Data Interface Numeric Array Channel
FP	Feed Pump
BMP	Ball Mill Pump
CRP	Cyclone Recycle Pump
CUFP	Cyclone Underflow Pump
FDP	Feed Disturbance Pump
NTP	Needle Tank Underflow Pump
PP	Product Pump

1.0 Purpose

During the 2022 Murdoch University Pilot Plant control system revision, every effort was made to ensure no functionality would be lost in the transition from the previous version which was introduced in 2012. However, it was decided relatively early in the rollout that the system should remain in a state, such that reversion to the previous system remains possible in the future. By making the reversion process as easy as possible, the intent is that should it be discovered that any required functionality is lost, or it be desired to revert to the old revision for any reason, the process can be carried out in a pain free manner.

The purpose of this document is to outline the process of upgrading from the 2012 to 2022 MUPP revision, and the process of downgrading from the 2022 to 2012 MUPP revision. This document outlines the necessary control module (CM) changes that must be carried out, and the procedure for carrying out these changes.

2.0 Scope

This document gives an account of all CMs that must be changed in the process of upgrading or downgrading the system. As several the CMs also have associated I/O changes that must take place, the document also describes the appropriate procedures pertaining to said I/O changes. As there are other situations in which an I/O change may be required to take place, this document can also be consulted for the methodology of making either a MODBUS/TCP or Analog I/O change; however, this is only a secondary function of this document.

If these documents are used, it assumed that all CMs remain in the Experion EPKS system. If this is not the case, locate the programs in the 'PP_Project' file of the Murdoch Engineering shared network. This document also does not cover the process of reverting the HMI to the previous revision; however, earlier versions of the HMI are also available 'PP_Project' file of the Murdoch Engineering shared network.

NOTE: To quickly view previous revisions of the HMI in the EPKS station environment, open the revision in the EPKS HMI development software and select 'View in Station' from the 'File' drop-down menu.

3.0 Procedure

The following sub-sections outline the procedure for upgrading to and downgrading from the MUPP 2022 control system revision.

3.1 Upgrading To 2022 Revision

When performing a MUPP upgrade to 2022 revision, a total of thirty-five control modules (CMs) must be deactivated, and fifty-two CMs must be activated (Table 2). The CMs correspond to seventeen controllers, wherein each controller:

- Two CMs must be deactivated, and one I/O channel must be unassigned.
- Three CMs must be activated, and one I/O channel must be assigned.

When performing an I/O change there are two different methodologies dependent on the I/O type. The first method is required for MODBUS/TCP equipment, with the second method being required for analog instrumentation; the following two subsections outline these methodologies.

Table 2: Control Modules that must be changed for an upgrade from 2012 to 2022 MUPP revision. Forth column indicates if an I/O change is required for the CM switch.

Associated Asset	ACTIVATE	DEACTIVATE	I/O CHANGE Required?
All	PID_PvSp_ARRAY-R0A	PID_PV_ARRAY	FALSE
FP-141	FP_REF_141-R1A	FP_REF_141	TRUE - MODBUS Out
	PID1_PV_SELECT-R1A	PID1_PV_SELECT	FALSE
	PID1_AutVar_SELECT-R0A		FALSE
BMP-241	BMP_REF_241-R1A	BMP_REF_241	TRUE - MODBUS Out
	PID3_PV_SELECT-R1A	PID3_PV_SELECT	FALSE
	PID3_AutVar_SELECT-R0A		FALSE
CRP-341	CRP_REF_341-R1A	CRP_REF_341	TRUE - MODBUS Out
	PID4_PV_SELECT-R1A	PID4_PV_SELECT	FALSE
	PID4_AutVar_SELECT-R0A		FALSE
CUFP-361	CUFP_REF_361-R1A	CUP_REF_361	TRUE - MODBUS Out
	PID5_PV_SELECT-R1A	PID5_PV_SELECT	FALSE
	PID5_AutVar_SELECT-R0A		FALSE
LUP-421	LUP_REF_421-R1A	LUP_REF_421	TRUE - MODBUS Out
	PID6_PV_SELECT-R1A	PID6_PV_SELECT	FALSE
	PID6_AutVar_SELECT-R0A		FALSE
FDP-521	FDP_REF_521-R1A	FDP_REF_521	TRUE - MODBUS Out
	PID7_PV_SELECT-R1A	PID7_PV_SELECT	FALSE
	PID7_AutVar_SELECT-R0A		FALSE
FCV-541	FCV_541-R1A	FCV_541	TRUE - Analog Out

Associated Asset	ACTIVATE	DEACTIVATE	I/O CHANGE Required?
	PID8_PV_SELECT-R1A	PID8_PV_SELECT	FALSE
	PID8_AutVar_SELECT-R0A		FALSE
NTP-561	NTP_REF_561-R1A	NTP_REF_561	TRUE - MODBUS Out
	PID7_PV_SELECT-R1A	PID9_PV_SELECT	FALSE
	PID7_AutVar_SELECT-R0A		FALSE
FCV-570	FCV_570-R1A	FCV_570	TRUE - Analog Out
	PID10_PV_SELECT-R1A	PID10_PV_SELECT	FALSE
	PID10_AutVar_SELECT-R0A		FALSE
FCV-574	FCV_574-R1A	FCV_574	TRUE - Analog Out
	PID11_PV_SELECT-R1A	PID11_PV_SELECT	FALSE
	PID11_AutVar_SELECT-R0A		FALSE
FCV-622	FCV_622-R1A	FCV_622	TRUE - Analog Out
	PID12_PV_SELECT-R1A	PID12_PV_SELECT	FALSE
	PID12_AutVar_SELECT-R0A		FALSE
FCV-642	FCV_642-R1A	FCV_642	TRUE - Analog Out
	PID13_PV_SELECT-R1A	PID13_PV_SELECT	FALSE
	PID13_AutVar_SELECT-R0A		FALSE
FCV-662	FCV_662-R1A	FCV_662	TRUE - Analog Out
	PID14_PV_SELECT-R1A	PID14_PV_SELECT	FALSE
	PID14_AutVar_SELECT-R0A		FALSE
FCV-688	FCV_688-R1A	FCV_688	TRUE - Analog Out
	PID15_PV_SELECT-R1A	PID15_PV_SELECT	FALSE
	PID15_AutVar_SELECT-R0A		FALSE
FCV-690	FCV_690-R1A	FCV_690	TRUE - Analog Out
	PID16_PV_SELECT-R1A	PID16_PV_SELECT	FALSE
	PID16_AutVar_SELECT-R0A		FALSE
PP-681	PP_REF_681-R1A	PP_REF_681	TRUE - MODBUS Out
	PID17_PV_SELECT-R1A	PID17_PV_SELECT	FALSE
	PID17_AutVar_SELECT-R0A		FALSE
FCV-571	FCV_571-R1A	FCV_571	TRUE - Analog Out
	PID19_PV_SELECT-R1A	PID19_PV_SELECT	FALSE
	PID19_AutVar_SELECT-R0A		FALSE

3.1.1 MODBUS/TCP I/O Changes

NOTE: The following procedure will use a single asset as an example (*LUP-421*); however, the procedure is identical for all assets requiring a MODBUS/TCP I/O change.

STEP 1: Deactivate 2012 Revision CM

1. Right-click CM.
2. Select 'selected item(s)' in the drop-down menu in the inactivate sub-menu (*Figure 1*).
3. Select 'Yes' in the popup menu.

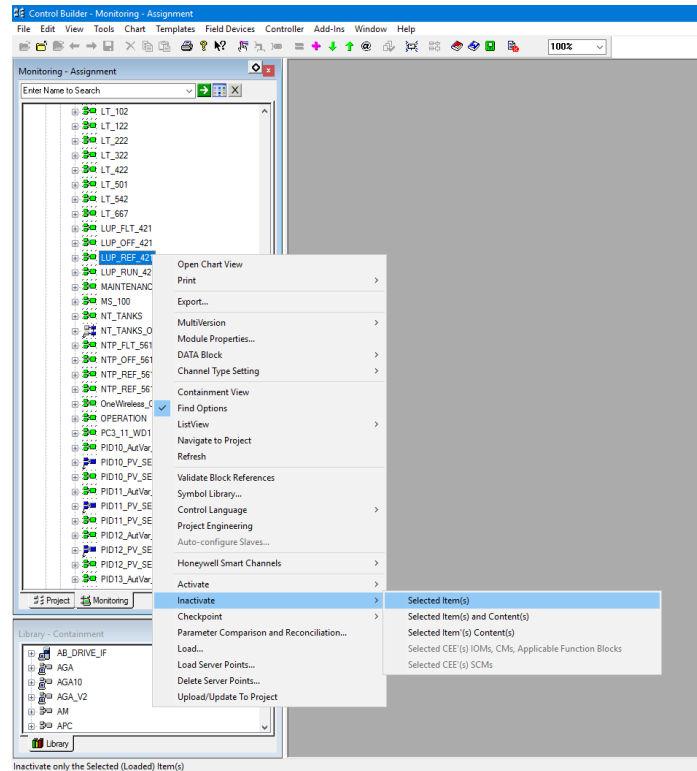


Figure 1: Deactivation of LUP_REF_421 control module (STEP 1a).

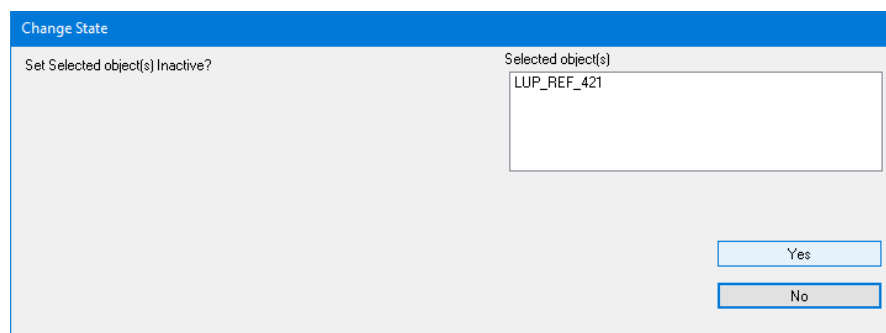


Figure 2: Deactivation of LUP_REF_421 control module (STEP 1b).

STEP 2: Delete 2012 Revision CM.

NOTE: This step is only required for CMs requiring an I/O change and is unnecessary for all CMs with 'FALSE' entered in the fourth column of Table 2.

1. Right-click CM.
2. Select 'Delete' from the drop-down menu.

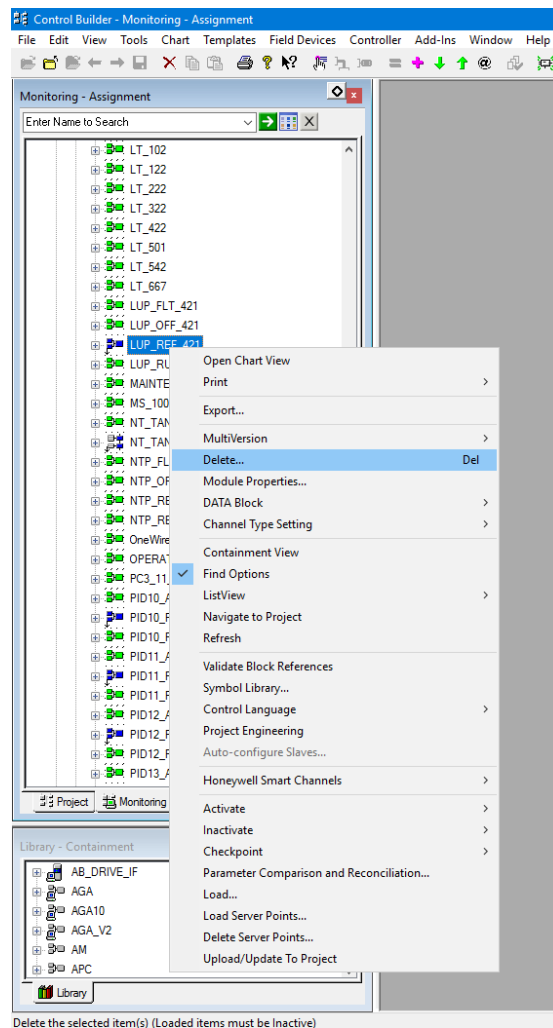


Figure 3: Deletion of LUP_REF_421 control module (Step 2).

STEP 3: Unassign Channel From 'PCDINMUARRCH' in 2012 Revision CM

NOTE: This step is only required for CMs requiring an I/O change and is unnecessary for all CMs with 'FALSE' entered in the fourth column of Table 2.

1. Locate the 'PCDINMUARRCH' block in CM (Figure 4).
NOTE: Take note of the device address (DEVADDR) on the front of the block graphic; in this case the device address is seven.
2. Select 'Unassign Request Block' from the 'parameters' popup menu in the 'Main' sub-menu (Figure 5).
3. Save and close CM.

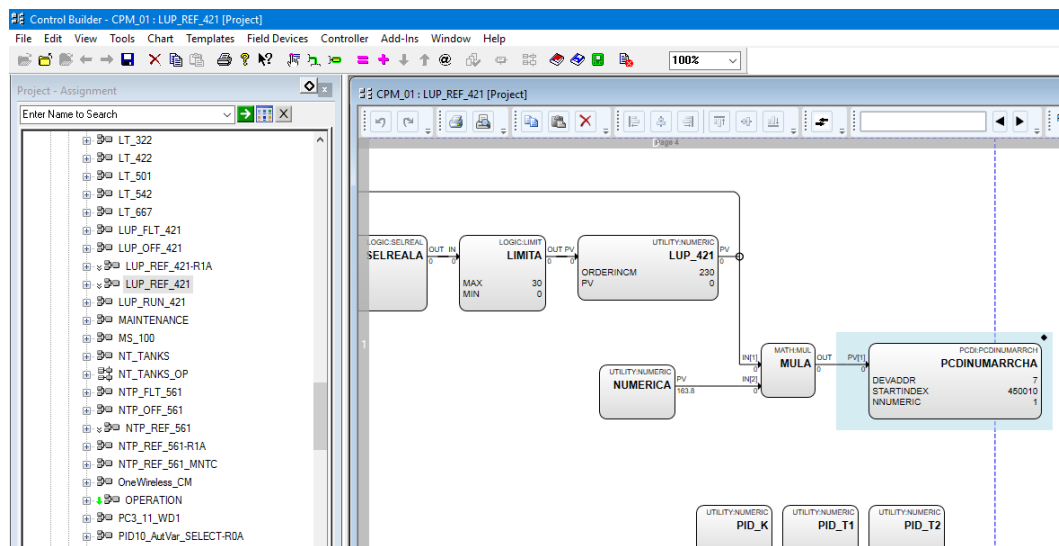


Figure 4: 'PCIDNUMARRCH' block in LUP_REF_421 (Step 3a), Take note of the device address (DEVADDR) on the front of the block graphic.

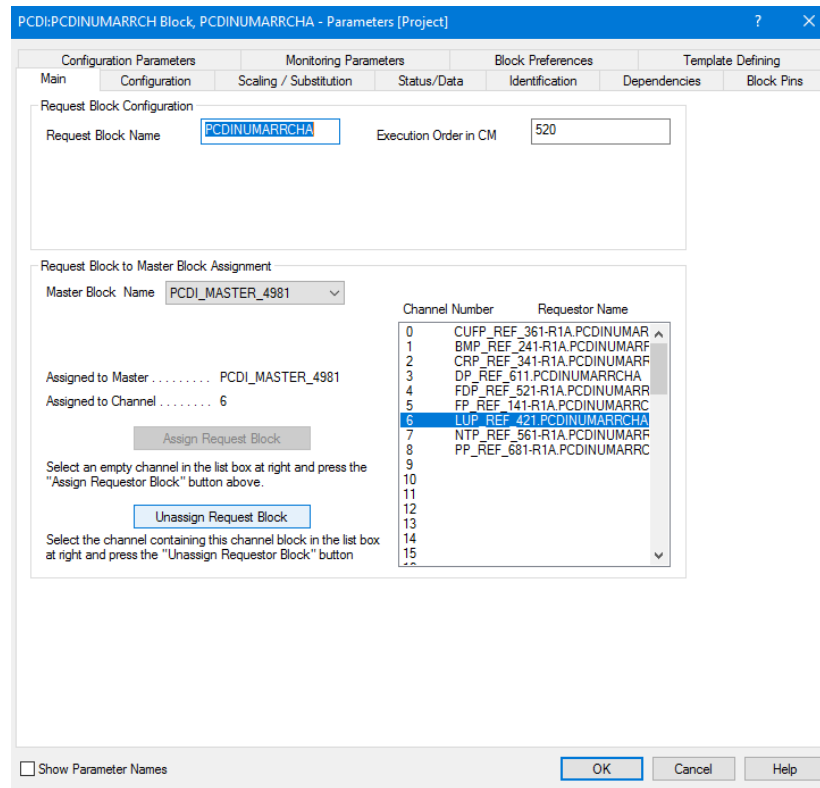


Figure 5: Unassigning LUP_REF_421.PCDINUMARRCH from output channel six (Step 3b).

STEP 4: Assign Channel To 'PCDINMUARRCH' in 2022 Revision CM

NOTE: This step is only required for CMs requiring an I/O change and is unnecessary for all CMs with 'FALSE' entered in the fourth column of Table 2.

1. Locate the 'PCDINMARRCH' block in CM.
NOTE: Take note of the device address (DEVADDR) on the front of the block graphic.
2. Select 'PCDI_MASTER_4981' in the Master Block Name parameter on the 'Main' sub-menu of the PCDINUMARRCH 'Parameters' pop-up menu.
3. *SELECTING THE SAME CHANNEL NUMBER AS THE PREVIOUSLY UNASSIGNED PARAMETER, select 'Assign Request Block' on the 'Main' sub-menu of the PCDINUMARRCH 'Parameters' pop-up menu (Figure 6).*
NOTE: Channel number and device address are NOT the same parameter and are typically different values.
4. Navigate to the 'Configuration' sub-menu of the PCDINUMARRCH 'Parameters' pop-up menu.
5. Change the device address to the address observed in step 3; in this case, we need to change this value from five to seven (Figure 7).
6. Save and close CM.

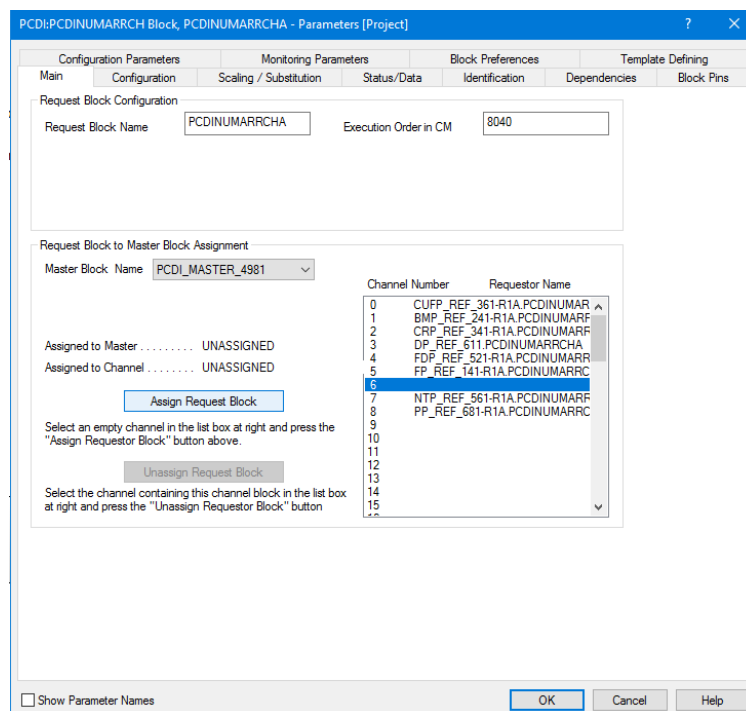


Figure 6: Assigning LUP_REF_421-R1A.PCIDNUMARRCH from output channel six (Step 4a).

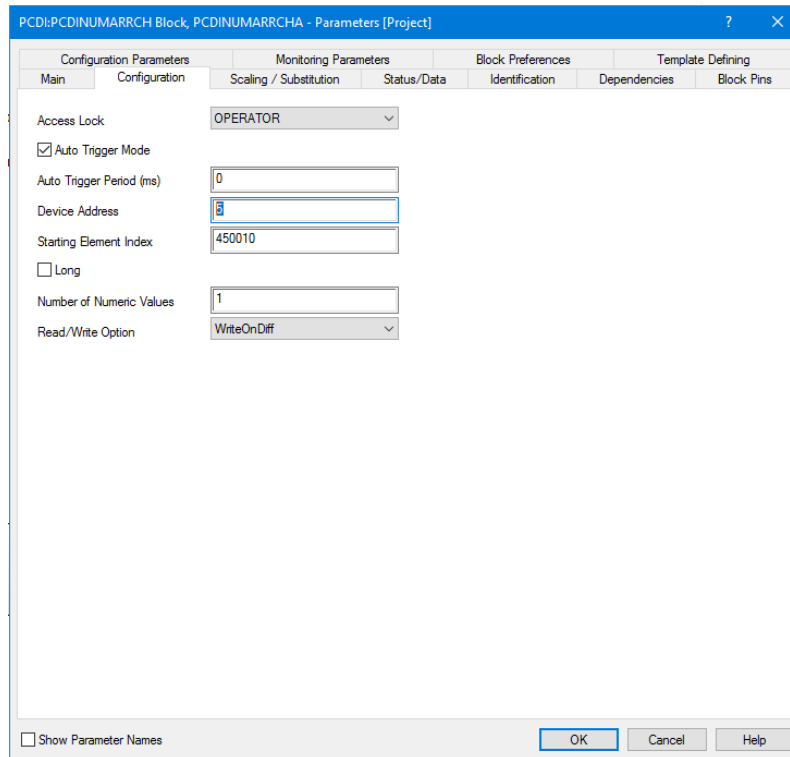


Figure 7: Changing Device address in LUP_REF_421-R1A.PCIDNUMARRCH (Step 4b).

STEP 5: Download 2022 Revision CM to C300 Controller.

At this stage, all procedures have been completed, and the CM and be downloaded to the C300 controller.

3.1.2 Analog I/O Changes

NOTE: The following procedure will use a single asset as an example (FCV-571); however, the procedure is identical for all assets requiring an analog I/O change.

STEP 1: Deactivate Analog Channel and Associated CM

NOTE: Deactivation of analog channels is only applicable when deactivating CMs routed to an analog I/O channel.

1. Deactivate the relevant analog channel from the C01L01_AO_07 module in IOLINK_01_01.
2. Deactivate the corresponding CM.

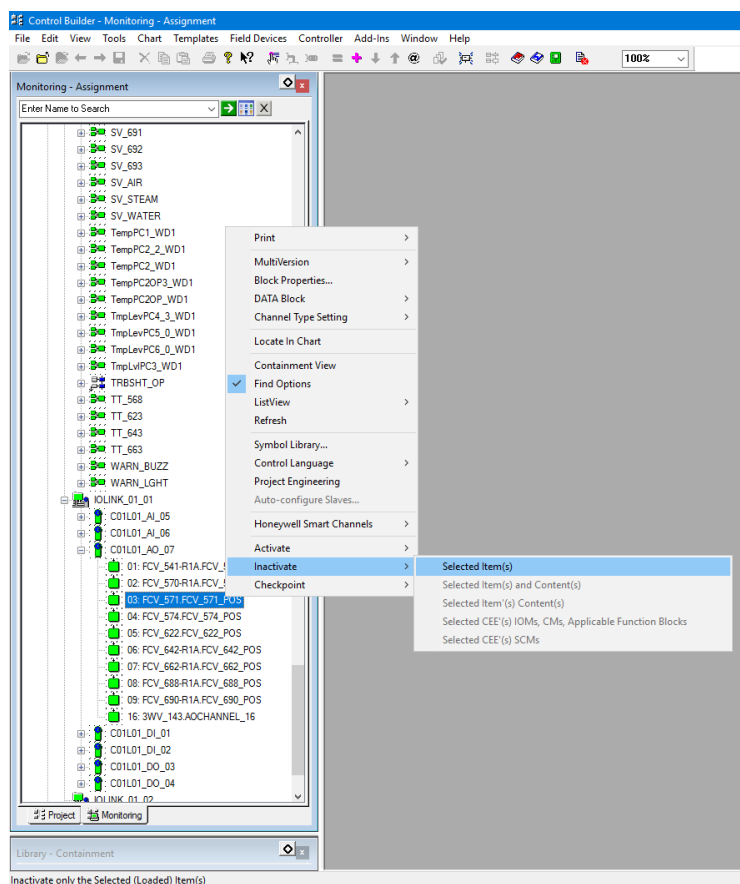


Figure 8: Deactivation of analog output channel three, associated with FCV-571 (STEP 1a).

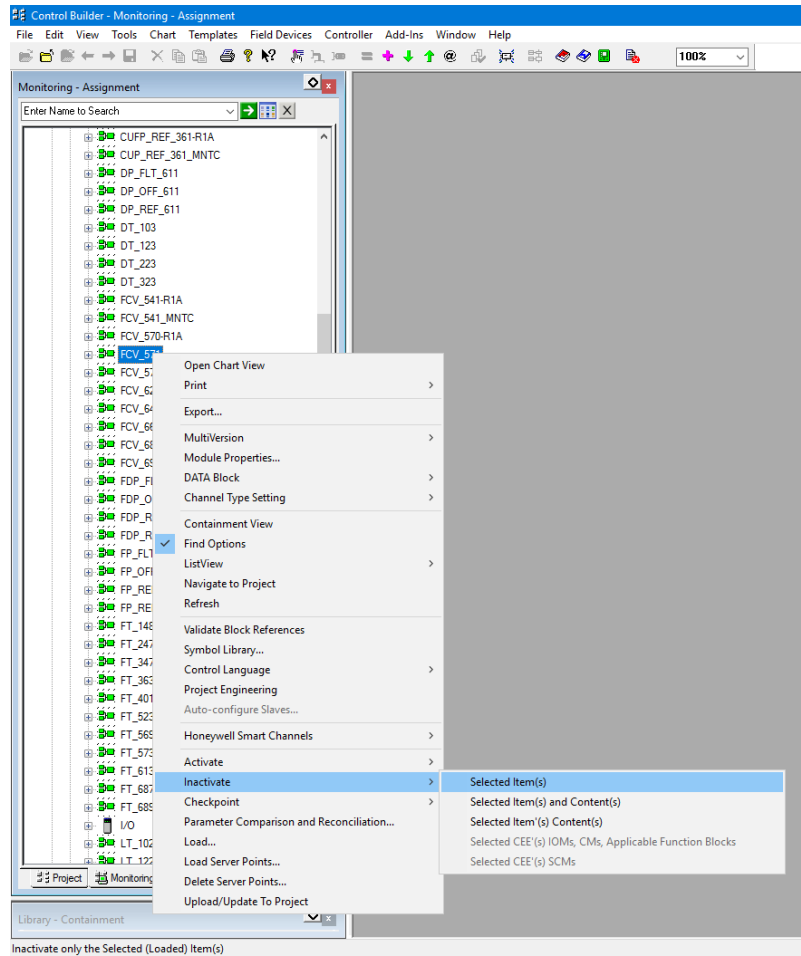


Figure 9: Deactivation of FCV_571 CM (Step 1b).

STEP 2: Delete 2012 Revision CM

NOTE: This step is only required for CMs requiring an I/O change and is unnecessary for all CMs with 'FALSE' entered in the fourth column of Table 2.

1. Right-click CM.
2. Select 'Delete' from the drop-down menu (*Figure 10*).

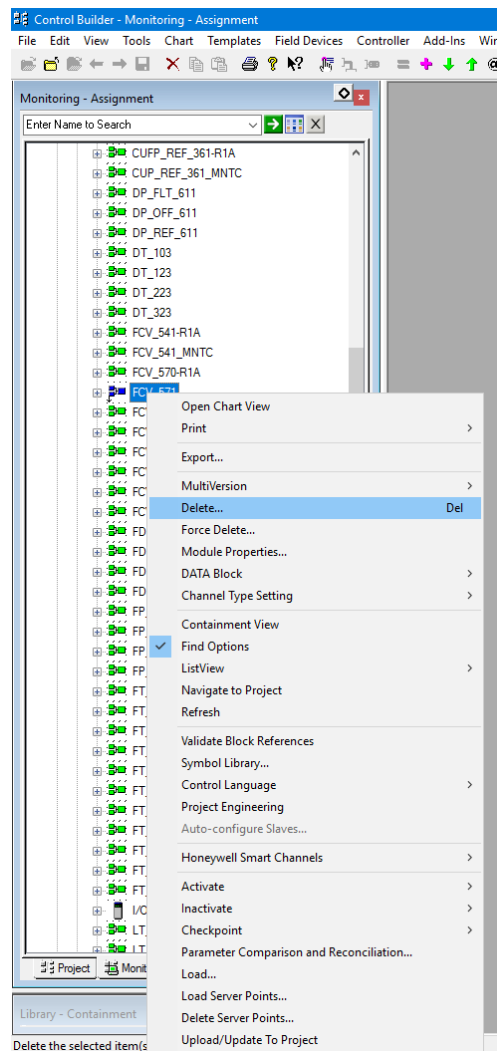


Figure 10: Deleting FCV_571 CM (Step 2).

STEP 3: Unassign Channel From 'SERIES_C_AOCHANNEL' in 2012 Revision CM

NOTE: This step is only required for CMs requiring an I/O change and is unnecessary for all CMs with 'FALSE' entered in the fourth column of Table 2.

1. Locate the SERIES_C_AOCHANNEL block.

NOTE: Take note of the channel number (CHANNUM) on the front of the block graphic; in this case, the channel number is three.

2. Right-click the SERIES_C_IO:AOCHANNEL block.
3. Select 'Function Block Unassign' from the drop-down menu (Figure 11).
4. Save and close CM.

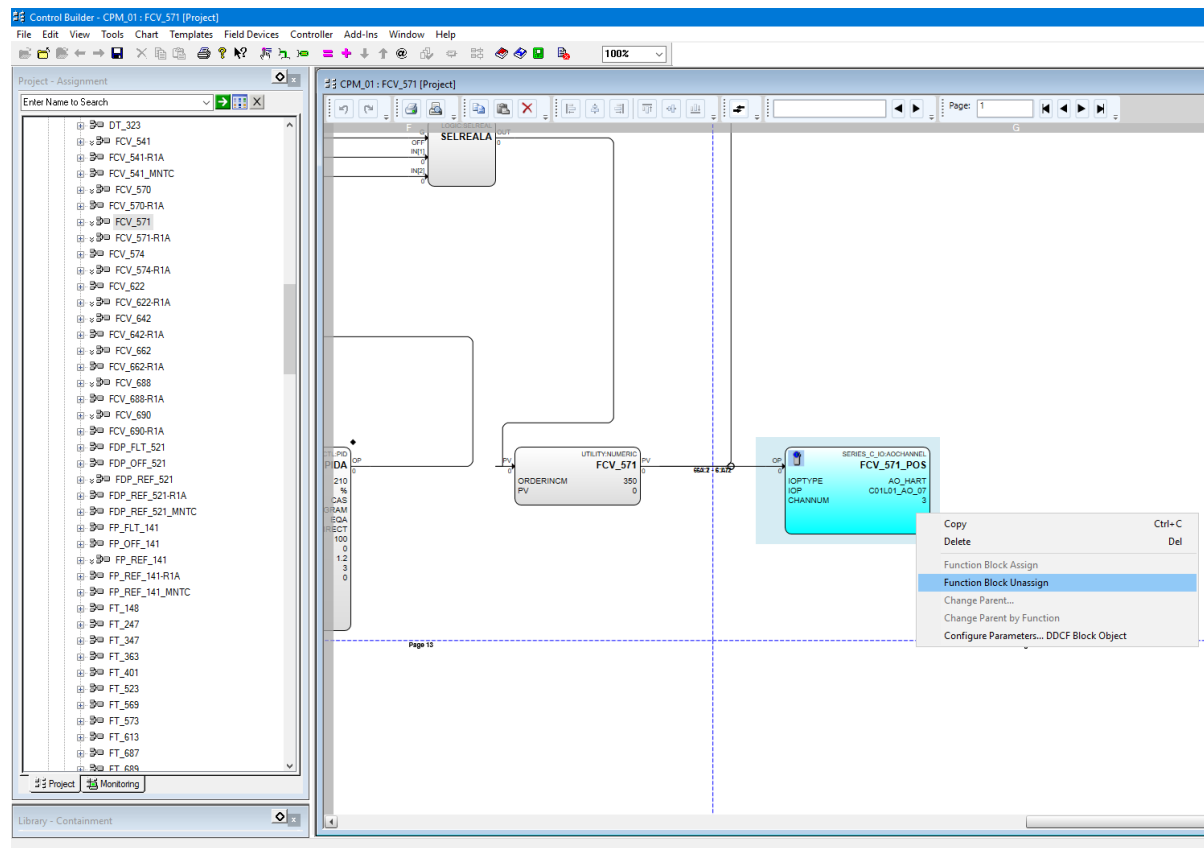


Figure 11: Unassigning analog output channel (Step 3).

STEP 4: Assign Channel to 'SERIES_C_AOCHANNEL' in 2022 Revision CM

NOTE: This step is only required for CMs requiring an I/O change and is unnecessary for all CMs with 'FALSE' entered in the fourth column of Table 2.

1. Locate the SERIES_C_AOCHANNEL block.
2. Right-click the SERIES_C_AOCHANNEL block.
3. Select 'Function Block Assign' from the drop-down menu.
4. Select analog output with channel number corresponding to the channel number observed in step 3.
5. Save and close CM.

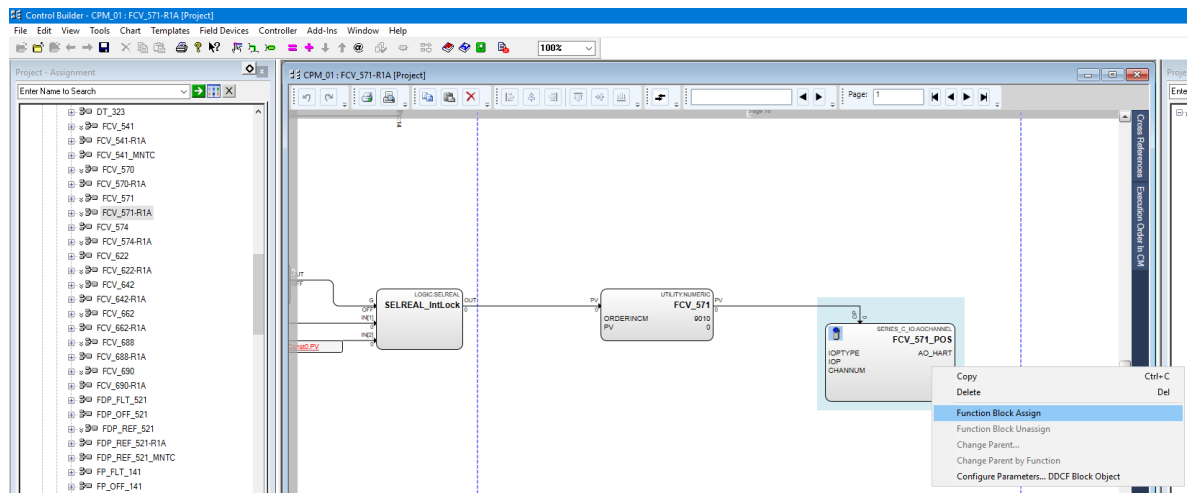


Figure 12: Assigning analog output channel (Step 4a).

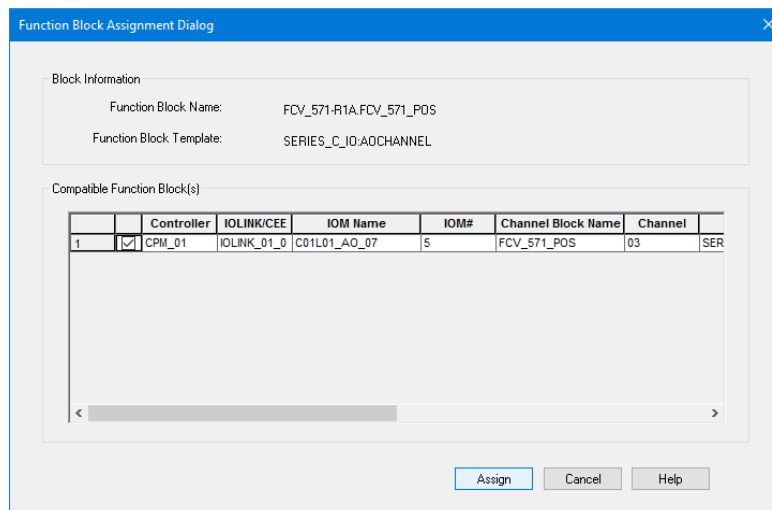


Figure 13: Assigning analog output channel (Step 4b).

STEP 5: Download 2022 Revision CM to C300 Controller.

At this stage all procedures have been completed and the CM and be downloaded to the C300 controller.

3.2 Downgrading from 2022 Revision

When performing a MUPP downgrade from the 2022 to 2012 revision, a total of fifty-two CMs must be deactivated and a total of thirty-five CMs must be activated (*Table 3*). The CMs correspond to seventeen controllers, wherein each controller:

- Three CMs must be deactivated, and one I/O channel must be unassigned.
- Two CMs must be activated, and one I/O channel must be assigned.

The procedure for activating CMs, deactivating CMs, and changing I/O channels is as described in the preceding section, *3.1 Upgrading To 2022 Revision*. The only difference in procedure is the CMs to be activated and deactivated; consult Table 3 for this information.

Table 3: Control Modules that must be changed for a downgrade from 2022 to 2012 MUPP revision. Forth column indicates if an I/O change is required for the CM switch.

Associated Asset	ACTIVATE	DEACTIVATE	I/O CHANGE Required?
All	PID_PV_ARRAY	PID_PvSp_ARRAY-R0A	FALSE
FP-141	FP_REF_141	FP_REF_141-R1A	TRUE - MODBUS Out
	PID1_PV_SELECT	PID1_PV_SELECT-R1A	FALSE
		PID1_AutVar_SELECT-R0A	FALSE
BMP-241	BMP_REF_241	BMP_REF_241-R1A	TRUE - MODBUS Out
	PID3_PV_SELECT	PID3_PV_SELECT-R1A	FALSE
		PID3_AutVar_SELECT-R0A	FALSE
CRP-341	CRP_REF_341	CRP_REF_341-R1A	TRUE - MODBUS Out
	PID4_PV_SELECT	PID4_PV_SELECT-R1A	FALSE
		PID4_AutVar_SELECT-R0A	FALSE
CUFP-361	CUP_REF_361	CUFP_REF_361-R1A	TRUE - MODBUS Out
	PID5_PV_SELECT	PID5_PV_SELECT-R1A	FALSE
		PID5_AutVar_SELECT-R0A	FALSE
LUP-421	LUP_REF_421	LUP_REF_421-R1A	TRUE - MODBUS Out
	PID6_PV_SELECT	PID6_PV_SELECT-R1A	FALSE
		PID6_AutVar_SELECT-R0A	FALSE
FDP-521	FDP_REF_521	FDP_REF_521-R1A	TRUE - MODBUS Out
	PID7_PV_SELECT	PID7_PV_SELECT-R1A	FALSE
		PID7_AutVar_SELECT-R0A	FALSE
FCV-541	FCV_541	FCV_541-R1A	TRUE - Analog Out
	PID8_PV_SELECT	PID8_PV_SELECT-R1A	FALSE
		PID8_AutVar_SELECT-R0A	FALSE
NTP-561	NTP_REF_561	NTP_REF_561-R1A	TRUE - MODBUS Out
	PID9_PV_SELECT	PID7_PV_SELECT-R1A	FALSE

Associated Asset	ACTIVATE	DEACTIVATE	I/O CHANGE Required?
		PID7_AutVar_SELECT-R0A	FALSE
FCV-570	FCV_570	FCV_570-R1A	TRUE - Analog Out
	PID10_PV_SELECT	PID10_PV_SELECT-R1A	FALSE
		PID10_AutVar_SELECT-R0A	FALSE
FCV-574	FCV_574	FCV_574-R1A	TRUE - Analog Out
	PID11_PV_SELECT	PID11_PV_SELECT-R1A	FALSE
		PID11_AutVar_SELECT-R0A	FALSE
FCV-622	FCV_622	FCV_622-R1A	TRUE - Analog Out
	PID12_PV_SELECT	PID12_PV_SELECT-R1A	FALSE
		PID12_AutVar_SELECT-R0A	FALSE
FCV-642	FCV_642	FCV_642-R1A	TRUE - Analog Out
	PID13_PV_SELECT	PID13_PV_SELECT-R1A	FALSE
		PID13_AutVar_SELECT-R0A	FALSE
FCV-662	FCV_662	FCV_662-R1A	TRUE - Analog Out
	PID14_PV_SELECT	PID14_PV_SELECT-R1A	FALSE
		PID14_AutVar_SELECT-R0A	FALSE
FCV-688	FCV_688	FCV_688-R1A	TRUE - Analog Out
	PID15_PV_SELECT	PID15_PV_SELECT-R1A	FALSE
		PID15_AutVar_SELECT-R0A	FALSE
FCV-690	FCV_690	FCV_690-R1A	TRUE - Analog Out
	PID16_PV_SELECT	PID16_PV_SELECT-R1A	FALSE
		PID16_AutVar_SELECT-R0A	FALSE
PP-681	PP_REF_681	PP_REF_681-R1A	TRUE - MODBUS Out
	PID17_PV_SELECT	PID17_PV_SELECT-R1A	FALSE
		PID17_AutVar_SELECT-R0A	FALSE
FCV-571	FCV_571	FCV_571-R1A	TRUE - Analog Out
	PID19_PV_SELECT	PID19_PV_SELECT-R1A	FALSE
		PID19_AutVar_SELECT-R0A	FALSE